Unit 2: Theoretical Structure of Quantum Information Systems

What is a qubit? Conceptual understanding using spin and polarization, Comparison: classical bits vs quantum bits, Quantum systems: trapped ions, superconducting circuits, photons (nanoengineering view), Quantum coherence and decoherence – intuitive explanation, Theoretical concepts: Hilbert spaces, quantum states, operators – only interpreted in abstract, The role of entanglement and non-locality in systems, Quantum information vs classical information: principles and differences, Philosophical implications: randomness, determinism, and observer role

What is a qubit? Conceptual understanding using spin and polarization

qubit (quantum bit) is the basic unit of quantum information, analogous to a classical bit. However, unlike a classical bit that can be either 0 or 1, a qubit can exist in a superposition of both states at the same time. To build an intuitive understanding of a qubit, let's explore it through two physical systems: spin and polarization.

Qubit Conceptually

At its core, a qubit is a two-level quantum system. Mathematically, it is represented as:

- $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$
- α and β are complex numbers satisfying $|\alpha|^2+|\beta|^2=1$ representing probability amplitude
- When measured, the qubit gives either 0 or 1 with probabilities $|\alpha|^2$ and $|\beta|^2$

Example 1: Spin of an Electron

Think of electron spin as a tiny compass needle pointing in a quantum direction.

Spin-up ($|\uparrow\rangle$) → represents $|0\rangle$

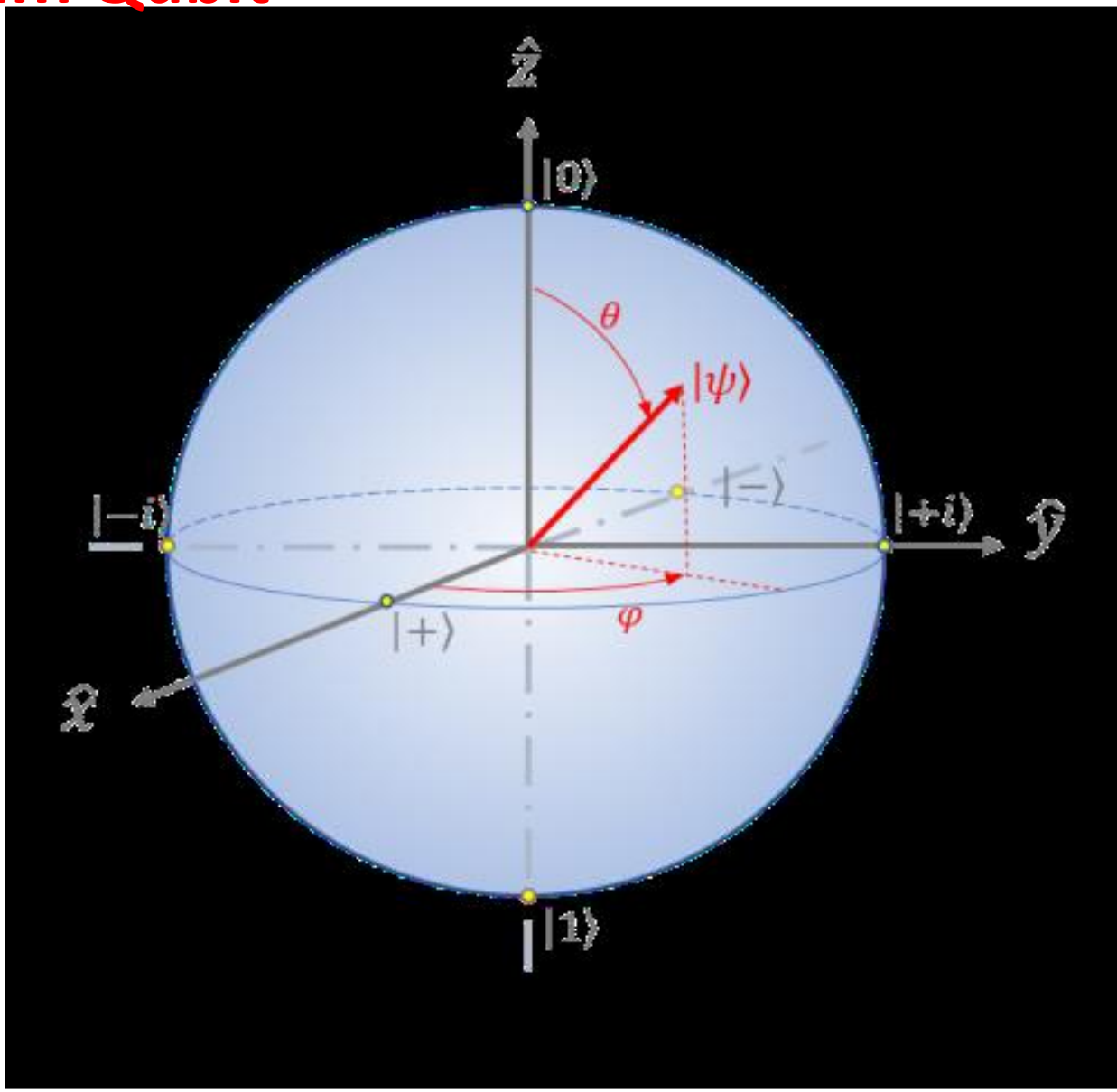
Spin-down ($|\downarrow\rangle$) → represents $|1\rangle$

But due to quantum mechanics, the spin can also point in any direction, not just up or down. This means the electron can be in a superposition.

$|\psi\rangle=\alpha|\uparrow\rangle+\beta|\downarrow\rangle$ Visually, imagine a sphere (called the Bloch Sphere), where any point on the surface represents a valid qubit state. Spin is not just binary—it can point anywhere on this sphere until it's measured.

Quantum Qubit

The Bloch sphere is often used to visually represent a qubit's state, where the poles correspond to |0⟩ and |1⟩, and any point on the sphere's surface represents a superposition.



The two-level superposition state corresponding to the point on the Bloch sphere is given by the general expression:

$$|\psi\rangle=\cos\left(\frac{\theta}{2}\right)|0\rangle+e^{i\varphi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

The two-level superposition state corresponding to the point $(\theta = \pi/2, \phi = \pi/4)$ on the Bloch sphere is given by the general expression:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

For $\theta = \pi/2$ and $\phi = \pi/4$, this becomes:

$$|\psi\rangle = \cos\left(\frac{\pi}{4}\right)|0\rangle + e^{i\pi/4}\sin\left(\frac{\pi}{4}\right)|1\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}e^{i\pi/4}|1\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + \frac{1+i}{\sqrt{2}}|1\rangle\right)$$

Therefore, the correct state is:

$$\frac{1}{\sqrt{2}}\left(|0\rangle + \frac{1+i}{\sqrt{2}}|1\rangle\right)$$

Example 2: Polarization of a Photon

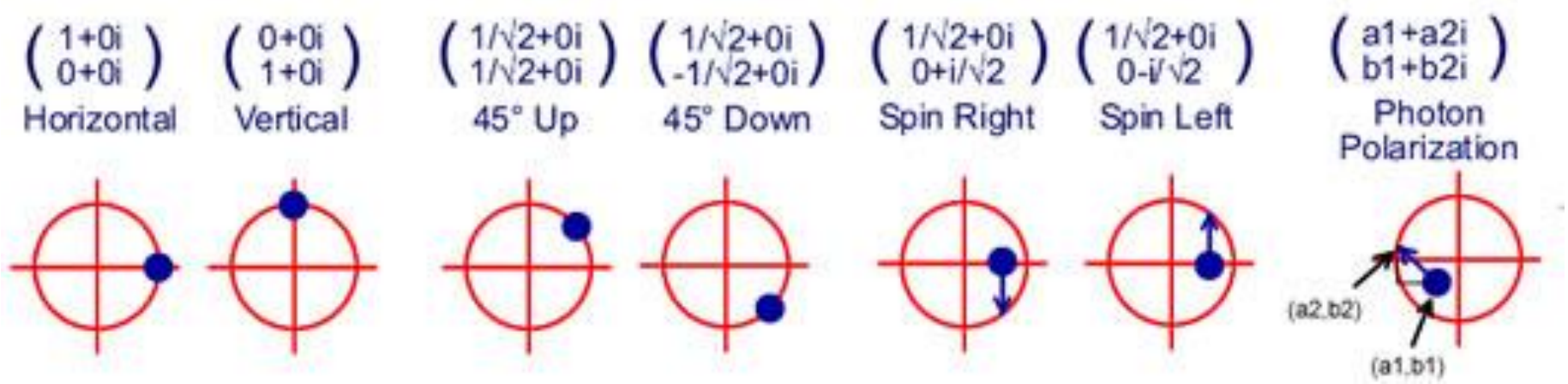
Photons (light particles) also behave like qubits using their polarization states.

Horizontal polarization → |0⟩

Vertical polarization → |1⟩

A photon can also be polarized at an angle (e.g., 45°), which is a superposition:

|ψ⟩=1/sqrt(2)(|horizontal⟩+|vertical⟩)



QUANTUM GATE

- A quantum gate is a fundamental operation in quantum computing that manipulates quantum bits or qubits.
- Quantum gates perform transformations on qubits using unitary matrices, enabling complex quantum computations through principles like superposition and entanglement.
- A unitary matrix is a complex square matrix U that satisfies the property:
- $UU^T = I$; where U^T is the conjugate transpose (also called Hermitian adjoint) of U, and I is the identity matrix. This means the inverse of the matrix is equal to its conjugate transpose, i.e., $U^T = U^{-1}$

- Quantum gates are reversible and are represented mathematically by unitary matrices—2x2 for single qubit gates, 4x4 for two-qubit gates, etc.
- Quantum gates are the building blocks for quantum circuits and algorithms, allowing quantum computers to solve certain problems much faster than classical computers.
- Examples include the Pauli-X gate (quantum NOT), Hadamard gate, and controlled gates like CNOT.
- Quantum gates work on qubits entangled with others and support quantum phenomena for powerful parallelism in computation.

FIDELITY

- Fidelity is the measurement of accuracy and stability of quantum states.
- High fidelity implies that quantum computation or communication is reliable with minimal errors, while low fidelity indicates higher noise or error effects.

Concept	Description
Fidelity	Accuracy and stability of quantum states
High Fidelity	Reliable operations, minimal errors/noise
Low Fidelity	Significant error, noise, or instability present

Quantum system:trapped ions, superconducting circuits, and photons

The terms **trapped ions**, **superconducting circuits**, and **photons** refer to three different **physical platforms** used to **realize qubits** in **quantum computing** and **quantum technologies**.

1. Trapped Ions

- **What it is:**
  Individual ions (charged atoms) trapped in electromagnetic fields using devices called **ion traps** (like Paul traps or Penning traps).
- **Qubits representation:**
  Internal electronic states of the ions (e.g., two energy levels) serve as the qubit states $|0\rangle$ and $|1\rangle$.
- **Quantum gate operations:**
  Controlled using lasers to manipulate the energy states and entangle the ions via their shared vibrational modes.

- **Advantages:**
  - Long **coherence times** (seconds to minutes).
  - High **fidelity** of gate operations.
  - Identical qubits – all ions are the same atom type.
- **Challenges:**
  - Slower gate speeds (milliseconds).
  - Scalability is complex due to control system overhead and ion crosstalk.
- **Example:**
  Companies like **IonQ** and **Quantinuum** use trapped ions.

ION TRAPS

- Ion traps are devices used to confine charged particles (ions) using electromagnetic fields, enabling precise control of individual ions. In quantum computing, ion traps are crucial for implementing qubits.
- Ion traps typically use a combination of radio-frequency (RF) and static electric fields (Paul traps) or strong magnetic and electric fields (Penning traps) to hold ions in a specific region.
- The ions are cooled to near absolute zero using laser cooling to reduce their motion, allowing quantum operations to be performed with high precision.

2. Superconducting Circuits

- **What it is:**
  Circuits made of superconducting materials (aluminium, niobium, etc) that operate at cryogenic temperatures (typically ~15 mK) using dilution refrigerators.
- **Qubits representation:**
  Use **Josephson junctions** to form artificial atoms, with two quantum energy levels used as $|0\rangle$ and $|1\rangle$.
- **Quantum gate operations:**
  Controlled by **microwave pulses** that excite and manipulate the qubit states.

- **Advantages:**
  - Fast gate speeds (nanoseconds).
  - Compatible with standard fabrication techniques (scalable).
  - Easy integration with control electronics.
- **Challenges:**
  - Shorter **coherence times** (microseconds to milliseconds).
  - Sensitive to noise and material defects.
- **Example:**
  Used by **IBM**, **Google**, **Rigetti**, and **Intel**.

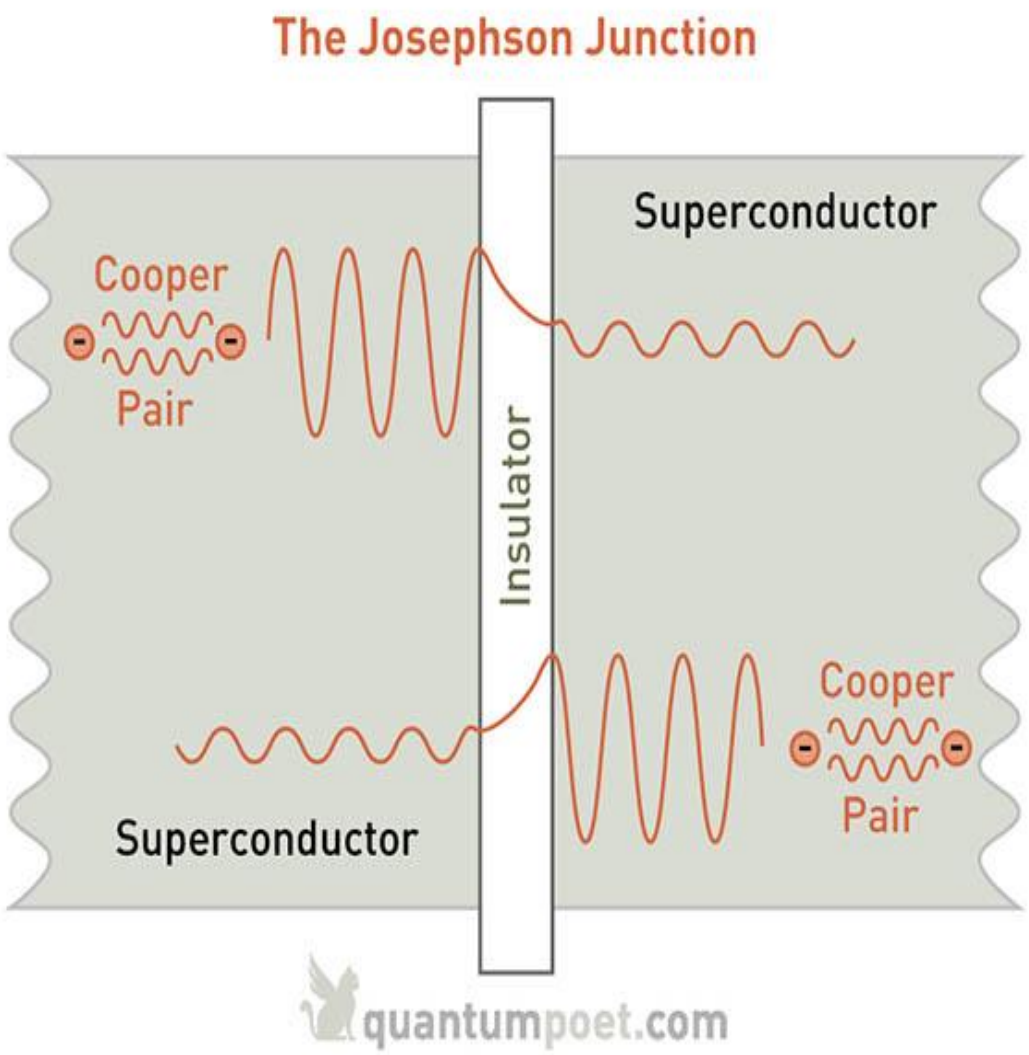
Josephson Junctions

- A crucial element in **Superconducting circuits** is the Josephson junction, which permits supercurrent to flow via quantum tunneling through an insulating barrier between superconductors.
- This junction introduces nonlinearity needed to create energy level spacing in qubits, making it possible to isolate two quantum states for computation.
- Josephson junctions are critical components in quantum computing (as qubits), superconducting quantum interference devices (SQUIDs), ultra-sensitive magnetometers, voltage standards, and digital electronic circuits exploiting superconducting properties.

JOSEPHSON JUNCTIONS

A Josephson junction is a fundamental quantum device consisting of two superconducting materials separated by a thin insulating barrier. It allows the tunneling of Cooper pairs (pairs of electrons with opposite spins) between the superconductors without resistance, this phenomenon known as the Josephson effect.

They enable a supercurrent (current without voltage) to flow across the insulating barrier. The current through the junction depends on the quantum phase difference between the superconductors.



- Josephson junctions are the core components of superconducting qubits, widely used in quantum computers developed by companies like IBM, Google, and Rigetti.
- They enable the construction of various types of qubits, such as charge qubits, flux qubits, phase qubits, and transmon qubits.

3. Photons

- **What it is:**
  Qubits are encoded in the quantum states of **light particles** (photons), such as polarization, path, or time-bin.
- **Qubits representation:**
  Polarization qubits use $|H\rangle$ (horizontal) and $|V\rangle$ (vertical) polarizations, for instance.
- **Quantum gate operations:**
  Performed using **beam splitters**, **phase shifters**, and **nonlinear optical elements** (sometimes aided by measurement and feed-forward).

- **Advantages:**
  - Photons travel at light speed – ideal for **quantum communication**.
  - Low interaction with the environment – long coherence lengths.
- **Challenges:**
  - Difficult to make **deterministic gates** (need probabilistic schemes).
  - Harder to **store** and **entangle** photons efficiently.
- **Example:**
  Used in **quantum communication**, **quantum networks**, and **linear optical quantum computing** (e.g., **Xanadu** in Canada).

Platform	Qubit Type	Gate Speed	Coherence Time	Strengths	Used By
Trapped Ions	Atomic ions	Slow	Very long (seconds)	High fidelity, long coherence	IonQ, Quantinuum
Superconducting	Josephson junctions	Fast	Medium (µs–ms)	Scalable, fast operations	IBM, Google, Rigetti
Photons	Light particles	Fast	Long (km-scale)	Ideal for communication	Xanadu, QKD systems

- Superconducting circuits: These are fabricated using superconducting materials to create circuits that behave quantum mechanically at very low temperatures. Major companies like IBM and Google use superconducting qubits for their scalable quantum processors. They offer relatively fast gate times and integration with existing semiconductor technologies.
- Trapped ions: Individual ions are confined and manipulated using electromagnetic fields in vacuum chambers. **IonQ** is a leading company leveraging trapped-ion qubits, which typically exhibit very long coherence times and high-fidelity operations, making them excellent for precise quantum computations.
- Photons: Quantum information is encoded in photons' properties such as polarization or path. PsiQuantum uses photonic qubits, which excel in low decoherence and connectivity over long distances, advantageous for quantum communication and scalable architectures.

Quantum coherence and decoherence

1. Quantum Coherence

What is it?

Coherence is the **"quantum harmony"** where a qubit maintains its delicate quantum state — such as a **superposition** (like being in both $|0\rangle$ and $|1\rangle$).

It means the system keeps its **phase relationships** intact — like keeping perfect rhythm and timing in a dance.

Intuition:

A **coherent qubit** can perform quantum interference, entanglement, and calculations.

It stays in its **quantum nature** and hasn't been disturbed by the outside world.

Analogy:

Quantum coherence is like a **group of violinists** playing in **perfect sync** — the music (quantum state) is beautiful and undisturbed.

2. Quantum Decoherence

What is it?

Decoherence happens when a quantum system interacts with its environment (noise, heat, stray particles, etc.).

This interaction **destroys** the quantum superposition — the qubit **loses its quantum behavior** and behaves more **classically**.

Intuition:

The system "forgets" its quantum state. It collapses from a smooth superposition to a definite state ($|0\rangle$ or $|1\rangle$).

This is **bad for quantum computers**, because it leads to **errors** and loss of information.

Analogy:

Decoherence is like one violinist in the orchestra suddenly **going out of tune** or being interrupted — the harmony is **lost**, and the performance falls apart.

Theoretical concepts: Hilbert spaces, quantum states, operators – only interpreted in abstract

Hilbert Spaces:-

1. What is a Hilbert Space?

A **Hilbert space** is a mathematical Vector space over complex numbers where:

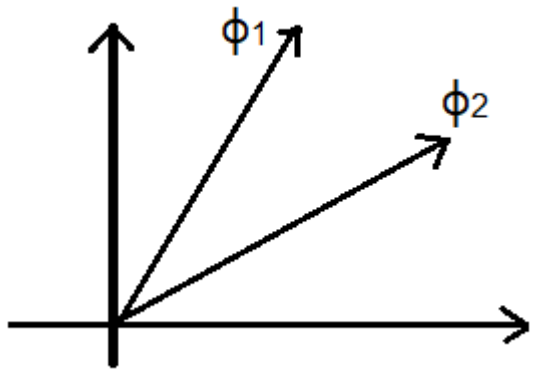
- Vectors represent **states** of a system (in quantum computing, quantum states of qubits).
- It's a **complete** vector space with an **inner product** (so we can measure angles and lengths between vectors).
- Orthonormal Basis are complete (no other Basis is required)
- In quantum computing:
- Each possible quantum state is a vector in a Hilbert space.
- The inner product lets us calculate **probabilities** of outcomes when measuring states.

- Hilbert Space is a linear Vector Space over complex numbers, whose inner product is defined. (satisfies the two conditions, Finite Dimensions and their Inner product is defined.

Quantum System is an element in the Hilbert Space.

Element is the State Vector belongs to Hilbert Space

Let us consider two basis states of Hilbert Space



Quantum System= $\psi = C_1\,\phi_1 + C_2\,\phi_2$

$= \sum_{i=1}^{2} C_i \;\; \phi_i$; Here C_i *is Complex* number

Any operator on ϕ_1 is a linear transform $\hat{A}\,\phi_1 = \lambda\,\phi_1$;

$\hat{A}$ is a Linear Operator $\hat{A}(a_1\,\phi_1 + a_2\,\phi_2) = a_1\hat{A}\;\phi_1 + a_2\hat{A}\,\phi_1$

Operation on Combined System is nothing but Sum of operation on Individual Systems

Hilbert space → A linear Vector space over Complex Numbers, whose Inner product is defined

Real space (3D)

$c \in \mathbb{R}$

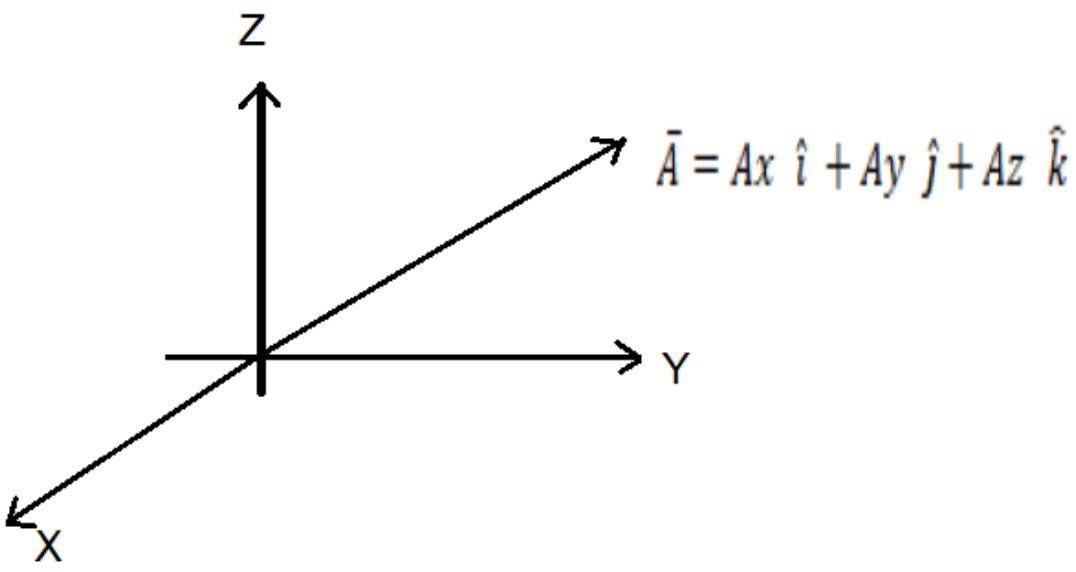
- $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$

Basis

$\{\hat{i}, \hat{j}, \hat{k}\}$

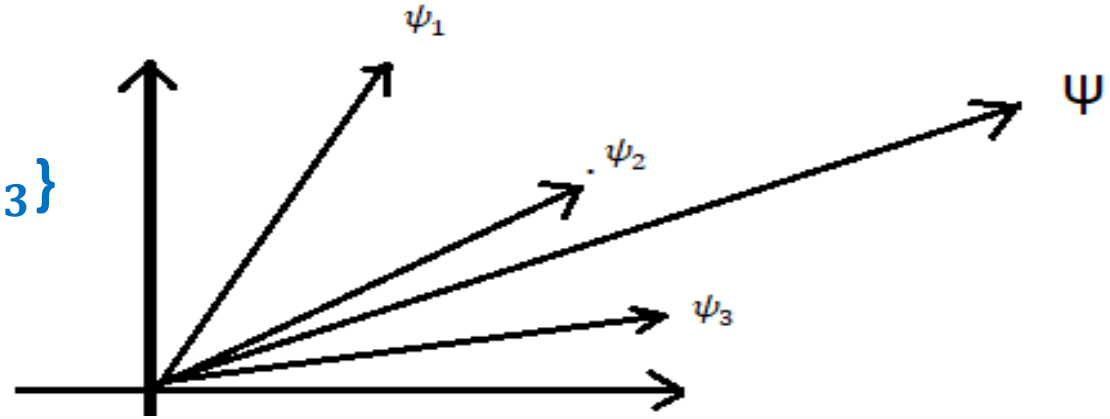
Basis States properties:

- $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ *(Normalized)*
- $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$ *(Orthogonal)*
- $\{\hat{i}, \hat{j}, \hat{k}\} \rightarrow$ Orthonormal condition
- Complete



Hilbert Space

Spanned by three **Basis States $\{\psi_1, \psi_2, \psi_3\}$**



Hilbert Space

- Hilbert space → A linear vector space over complex numbers whose inner product is defined.
- Spanned by $\{\psi_1, \psi_2, \psi_3, \psi_4\}$
- $\psi = c_1\psi_1 + c_2\psi_2 + c_3\psi_3 + c_4\psi_4$
- Basis: $\{\psi_1, \psi_2, \psi_3\}$
- Inner product relation (Orthogonality and Normalization):

$$\delta_{i,j} = 1;\ where\ i = j$$

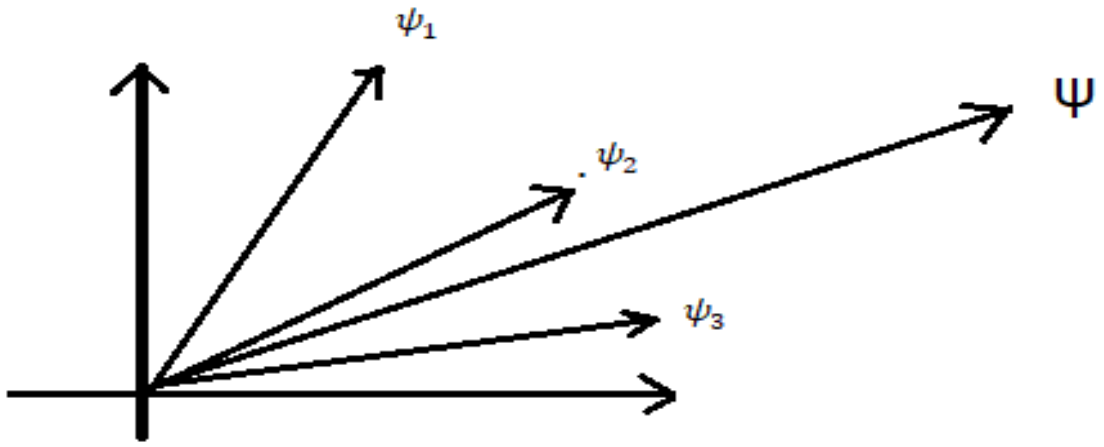
$$\delta_{i,j} = 0;\ where\ i \neq j$$

$$\int \psi_i^* \psi_j d\tau = \delta_{ij} \implies \langle \psi_i | \psi_j \rangle = \delta_{ij}$$

- Orthogonal and Normalized basis ⇒ Basis is **Complete**

- Hilbert space spanned by three **Basis States** {ψ1,ψ2,ψ3}

- Ψ=$c1\psi1+c2\psi2+c3\psi3$
- Basis {ψ1,ψ2,ψ3}
-
- The inner product relation
- Orthogonality & Superposition is given by



-
- ⟨ψi|ψj⟩=∫ψi∗ψj dv=δij
- ⇒⟨ψi|ψj⟩=δij
-

$$\delta_{i,j} = 1;\ where\ i = j$$

$$\delta_{i,j} = 0;\ where\ i \neq j$$

Orthogonal and Normalized basis means the Basis is **Complete**

Let $\psi \in \mathcal{H} \Rightarrow$ ψ is normalized?

$$\psi = c_1\psi_1 + c_2\psi_2 + c_3\psi_3$$

$$\int \psi^*\psi \, dv = \int (c_1^*\psi_1^* + c_2^*\psi_2^* + c_3^*\psi_3^*)(c_1\psi_1 + c_2\psi_2 + c_3\psi_3)dv$$

Expanding and integrating:

$$\int \psi^*\psi \, dv = |c_1|^2 \int \psi_1^*\psi_1 d\tau + |c_2|^2 \int \psi_2^*\psi_2 d\tau + |c_3|^2 \int \psi_3^*\psi_3 d\tau$$

Assuming orthonormality ($\int \psi_i^*\psi_i dv = 1$, and $\int \psi_i^*\psi_j dv = 0$ for $i \neq j$):

$$\int \psi^* \psi \, dv = |c_1|^2 + |c_2|^2 + |c_3|^2$$

If

$$\int \psi^* \psi dv = 1$$

then

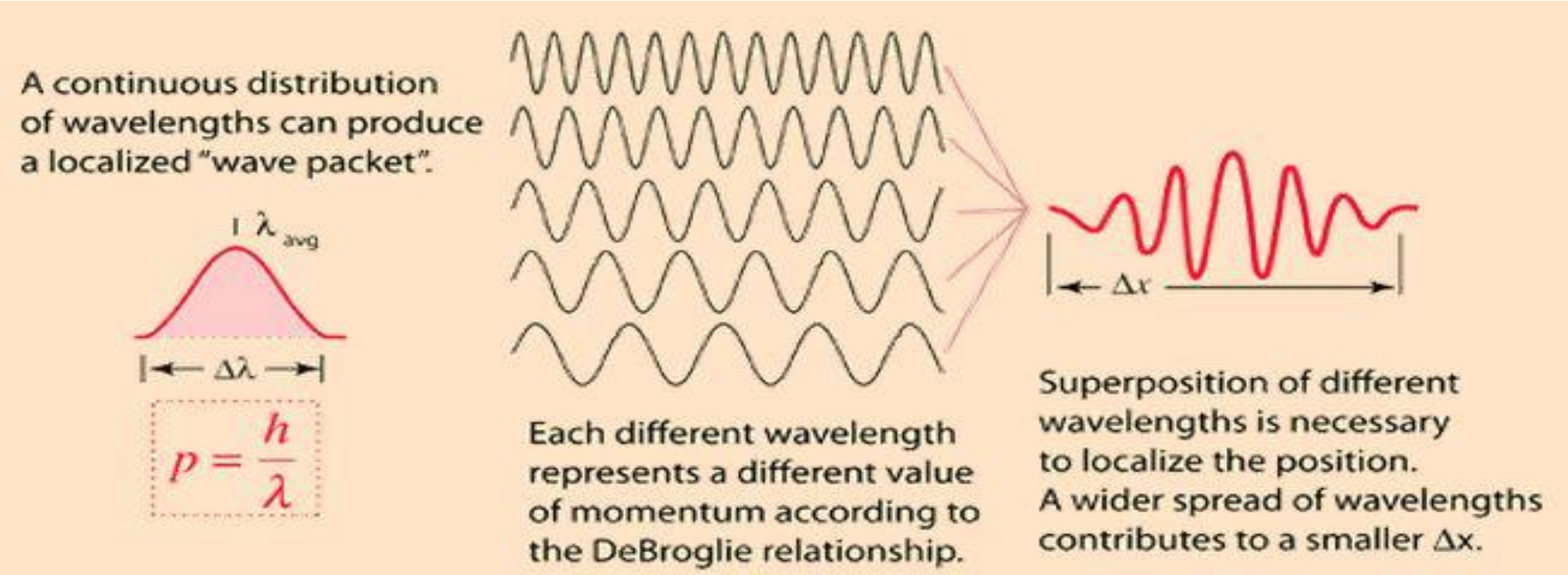
ψ is normalized.

Thus:

$$|c_1|^2 + |c_2|^2 + |c_3|^2 = 1$$

Sum of squares of amplitudes $= 1$.

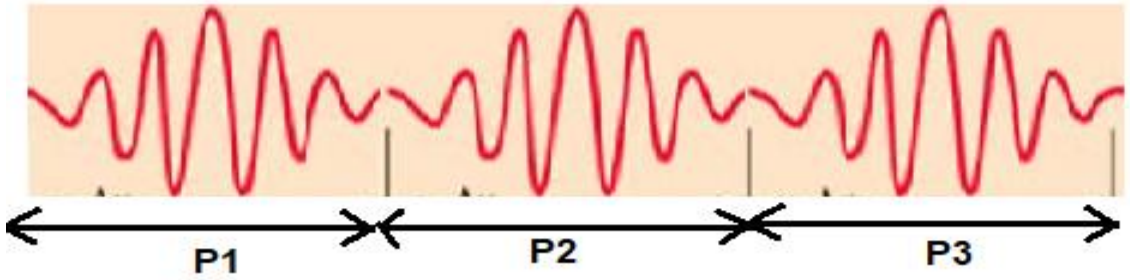
DE BROGLIE WAVEPACKET



Wave packet is nothing but Superposition of so many waves constructively where the particle is present and destructively where the particle is absent.

Constructive Interference and Destructive Interference

- Electron is moving with velocity - v is described by a wave packet
- .



2. How Qubits Live in Hilbert Space

Single qubit: Hilbert space is 2-dimensional, usually written as

$$H_1 = C^2$$

Any state $|\psi\rangle$ can be written as: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

where $\alpha, \beta \in C$ and $|\alpha|^2 + |\beta|^2 = 1$

This normalization comes from the inner product.

Two qubits: Hilbert space is 4-dimensional, usually written as

$$H_2 = C^2 \otimes C^2 = C^4$$

Basis: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

State example:

$|\psi\rangle = |00\rangle + |11\rangle / \mathrm{sqrt}(2)$ (entangled Bell state)

n qubits: Hilbert space dimension is 2^n

$H_n = C^{2 \otimes n}$

Quantum States: Vectors of Possibility

A quantum state is not just a single outcome — it's a **mathematical object (a vector)** that encodes *all possible* outcomes for a quantum system **and** the probabilities of each.

1. Why a Vector?

In quantum mechanics, the possible configurations of a system form a **Hilbert space**.

- A state $|\psi\rangle$ is a **vector** in that space.
- The *direction* of the vector encodes **how likely** different outcomes are.
- The *length* of the vector is always **1** (normalization rule).

2.The Possibility Part

For a single qubit:

$|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$

- $|0\rangle$ and $|1\rangle$ are **basis vectors** (like x and y axes for probability space).
- α and $\beta$are complex numbers whose squared magnitudes $\alpha^2+\beta^2$ are the **probabilities** of measuring 0 or 1.

Operators: Abstract Machines Acting on Quantum States

1. What They Are

In quantum mechanics, an **operator** is a **mathematical function** that takes one quantum state (a vector) and produces another quantum state.

Formally: $\hat{O}|\psi\rangle=|\phi\rangle$ or $\hat{O}|\psi\rangle\rightarrow|\phi\rangle$;

This means the operator $\hat{O}$ acts on the state $|\psi\rangle$ and transforms it into the state $|\phi\rangle$. This is the typical way to represent the action of an operator on a quantum state vector in quantum mechanics.

Why “Abstract Machines” Fits

- Each operator is like a machine:
  - **We input** a quantum state.
  - The machine processes it according to its internal rule.
  - You **output** the transformed state.
- Different operators = different machines (rotators, flippers, energy measurers, etc.).

Types of Operators

Unitary Operators – Machines that **evolve** quantum states without losing information.

Quantum gates in quantum computing (Hadamard, Pauli-X, etc.).

Preserve vector length (probabilities sum to 1).

Example: Hadamard transforms $|0\rangle$ into a 50–50 blend of $|0\rangle$ and $|1\rangle$.

Hermitian Operators – Machines that **measure** something.

- Their eigenvalues are real → correspond to possible measurement outcomes.
- Example: $\hat{Z}$ operator for measuring spin in the z-direction.

Projection Operators – Machines that **filter** states to certain subspaces.

Example: If we project onto $|0\rangle$, we keep only the "0" component.

If we project onto $|1\rangle$, we keep only the "1" component.

The role of entanglement and non-locality in systems

1. Entanglement: The Resource

Definition: Entanglement is a uniquely quantum correlation where the state of one particle is inseparably linked to the state of another, no matter the distance.

Mathematical form:

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle_A|1\rangle_B + |1\rangle_A|0\rangle_B\right)$$

- Neither particle has an independent state.
- The system is described only as a **whole**.

In this state, neither particle A nor B has a definite value of 0 or 1 until measured, but their outcomes are perfectly correlated: if one is found in state 0, the other will be in state 1, and vice versa.

The Roles of of Entanglement in Systems

Quantum Communication: Enables *quantum teleportation* and *superdense coding*.

Quantum Cryptography: Basis of secure key distribution (QKD, e.g., Ekert protocol).

Quantum Computing: Provides "quantum parallelism" and speedup in algorithms (e.g., Shor's factoring, Grover's search).

Quantum Sensing & Metrology: Entangled particles enhance precision beyond classical limits (Heisenberg limit).

2. Non-Locality: Beyond Classical Correlations

Definition: Non-locality refers to correlations between entangled particles that cannot be explained by classical local hidden variables.

Demonstrated through **Bell's Theorem**: quantum predictions violate Bell inequalities, showing that nature is not governed by purely local classical rules.

Example: Measuring spin of one electron instantly influences the statistics of another entangled electron, even light-years apart.

The Roles of of Non-locality in Systems

Foundational: Non-locality confirms quantum theory's departure from classical realism.

Quantum Networks: Non-local correlations enable distributed quantum computing and entanglement swapping.

Security Guarantee: Device-independent quantum cryptography uses Bell-inequality violations to prove security *without trusting devices*.

- **Local:** Things are influenced only by their immediate surroundings. If one thing changes, its effect travels through space gradually, no faster than the speed of light. For example, if you flip a switch, the electricity flows through wires to a bulb; nothing happens instantly at a distance. All classical physics follows this rule.
- **Non-local:** Changes in one place can instantly affect something far away, with no delay and no need for anything to travel between them. In quantum physics, "non-local" often describes the strange connection called *entanglement:* measurement of one particle can immediately relate to another, even if they're miles apart.

3. How They Work Together

Entanglement = the phenomenon (shared quantum state).

Non-locality = the manifestation (observed correlations defying classical explanations).

In real systems:

- **Entanglement provides the *resource*.**
- **Non-locality ensures the *advantage over classical systems*.**

4. Examples

1.Einstein-Podolsky-Rosen (EPR) Experiment:

1. Two entangled photons measured at distant labs show correlated results stronger than classical limits.

2.Quantum Teleportation:

1. Entanglement + classical communication allows transmission of an unknown quantum state without physically sending the particle.

3.Bell Test Experiments (Aspect, Zeilinger, etc.):

1. Verified non-locality experimentally, confirming quantum mechanics' prediction

QUANTUM INFORMATION VS CLASSICAL INFORMATION: PRINCIPLES AND DIFFERENCES

Classical Information Principles

1. **Definiteness** – A classical bit is always in a definite state (0 or 1).
2. **Determinism** – Processing follows deterministic logic (AND, OR, NOT, etc.).
3. **Copyability** – Information can be freely duplicated without change.
4. **Measurement Transparency** – Reading data does not disturb or alter it.
5. **Error Correction** – Redundancy-based codes (e.g., parity bits, checksums) correct errors.

Quantum Information Principles

- **Superposition Principle** – A qubit exists in a linear combination of states:
- $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$ where $|\alpha|2+|\beta|2=1$.
- **Entanglement Principle** – Qubits can share correlations beyond classical probability.
- **Measurement Principle** – Measurement collapses the superposition into one definite outcome.
- **No-Cloning Principle** – Unknown quantum states cannot be copied exactly.
- **Unitarity & Reversibility** – Quantum evolution is governed by unitary transformations (reversible).
- **Decoherence Principle** – Interaction with environment destroys quantum information.

ASPECT	CLASSICAL INFORMATION	QUANTUM INFORMATION
Basic Unit	Bit (0 or 1)	Qubit (superposition of 0 and 1)
State Representation	Binary strings	Quantum states (vectors in Hilbert space)
Storage	n bits = one definite configuration	n qubits = superposition of 2^n configurations
Processing	Boolean logic, deterministic	Quantum gates (unitary), probabilistic outcomes
Measurement	Non-invasive	Collapses state, probabilistic
Correlation	Classical probability correlations	Quantum entanglement (non-local)
Copying	Allowed	Forbidden (No-Cloning Theorem)
Error Correction	Redundancy codes (parity, CRC, Hamming)	Quantum error correction (Shor, Steane codes)
Security	Based on computational hardness (e.g., RSA)	Based on quantum physics laws (e.g., QKD)
Applications	Computing, storage, communication	Quantum computing, cryptography, teleportation, sensing

Philosophical implications: randomness, determinism, and observer role

1. Randomness

- **Classical view:**
  - **Randomness is usually due to *incomplete knowledge*.**
  - **Example: A coin toss is deterministic in principle (laws of motion), but unpredictable in practice.**
- **Quantum view:**
  - **Randomness is intrinsic to nature.**
  - **When we measure a quantum state (e.g., a superposed qubit), the outcome is fundamentally probabilistic — not just ignorance.**

2. Determinism

- **Quantum evolution before measurement:**
  - Governed by Schrödinger's equation, which is deterministic and unitary.
  - If you know the state, you can predict its evolution perfectly.
- But upon measurement:
  - Determinism breaks → wavefunction collapse introduces irreducible randomness.

3. The Role of the Observer

- **Classical physics:**
  - The observer is passive; measurement just reveals a pre-existing property.
- **Quantum physics:**
  - The observer is active; measurement changes the state (collapse).
  - Example: Before measuring an electron's spin, it has no definite spin direction. The act of observation gives it one.
- **Philosophical debates:**
  - Copenhagen Interpretation: Observer causes collapse; reality is not definite until measured.
  - Many-Worlds Interpretation: No collapse — all outcomes happen in branching universes.
  - QBism & Relational QM: Quantum states are about *observer's knowledge/relations*, not objective reality.

4. Entanglement, Non-locality & Philosophy

- Entanglement shows that reality is **holistic** — parts don't have independent existence, only the whole system does.
- Non-locality challenges classical ideas of **separability and locality**: effects appear correlated across space without signals.
- Raises questions:
  - Is the universe fundamentally interconnected?
  - Do space and time limit reality, or just our classical intuition?